

Multidrone Parcel Delivery via Public Vehicles: A Joint Optimization Approach

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Abstract—As one of the promising self-powered sensors on Internet of Things (IoT) platforms, unmanned aerial vehicles (UAVs) have attracted much attention for parcel delivery. Their high flexibility and low cost facilitate last-one-mile delivery. However, the limitations of battery capacity and payloads prevent drones from delivering independently over large scales. In this case, it is available to employ vehicles to assist the drones. The vehicles can be private-owned trucks and vehicles in public transportation systems (PTSs). Compared to trucks, PTSs, such as buses and trains, do not require extra operating and fuel costs. Given these advantages, this article adopts PTSs to assist UAVs in parcel delivery. Nevertheless, the fixed routes and schedules of public vehicles pose new challenges to the routing and scheduling problem for PTS-assisted multidrone parcel delivery (RSPMD). To tackle the problem, we propose a novel routing and scheduling algorithm, referred to as the PTS-assisted multidrone parcel delivery (PDD) algorithm. Considering the schedules of the public vehicles, the algorithm jointly optimizes the distance and time cost of drones by iteratively combining parts of existing routes. To the best of our knowledge, we are the first to address RSPMD in which UAVs ride public vehicles to deliver parcels in a wide area. Simulation results are finally presented to demonstrate that PDD outperforms existing solutions in terms of effectiveness and efficiency.

Index Terms—Intelligent transportation systems, smart cities, unmanned aerial vehicle (UAV), vehicle routing problem.

I. INTRODUCTION

IN RECENT years, unmanned aerial vehicles (UAVs), especially drones, have achieved great success in various Internet of Things (IoT) domains, such as remote inspection [1], [2], data collection [3], [4], environment monitoring [5], [6], and so on. Due to their agility, convenience, fast moving, and low costs, logistics companies, such as Amazon and DHL, have also exploited UAVs to implement smart last-one-mile parcel delivery [7].

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Nevertheless, the limited payload and battery capacity prevent UAVs from performing long-distance delivery missions [8]. To remedy the shortcoming, previous work used trucks to transport drones [9], [10], [11], [12], [13], which, however, incurs additional fuel costs and pollution. As a result, a few previous studies proposed to replace trucks with vehicles in public transportation systems (PTSs) to deliver parcels, such as trains [14]. The PTS-assisted UAV delivery approach leverages buses/trains to extend the delivery range of UAVs but incurs no extra operating or fuel costs.

The PTS-assisted UAV package delivery is a prospective technology which can enhance the efficiency and convenience of logistics. However, there are still some problems in the current research. Articles [14], [15], [16], and [17] only discussed the situation of a single-drone riding public transportation to deliver packages. Papers [14] and [15] suffered from the influence of the train schedule. The interval between train arrivals is long, resulting in considerable time waste for the drone. Furthermore, the customers who can get the service should be close to the train track. Choudhury et al. [18] established a multidrone delivery model, but without considering the constraints of drone capacity and payload, which are very important parameters for drone delivery. In contrast, we propose to use multiple drones riding buses.¹ Most importantly, we jointly optimize the time and distance costs of drones based on the constraints of drone payload and battery capacity to minimize the total delivery time.

To this end, we study the routing and scheduling problem for PTS-assisted multidrone parcel delivery (RSPMD) in which UAVs ride on public vehicles (e.g., trains and buses) to deliver parcels in a wide area. The problem utilizes multiple drones to serve customers in parallel, which is similar to a vehicle routing problem with multiple trips (VRPMTs) [19]. Nevertheless, VRPMT neglects fixed routes and schedules of buses. In addition, the routing model of RSPMD contains both drones and buses, while VRPMT uses only one type of vehicle. Given the above factors, it is challenging to solve this problem. First, we need to determine the public transportation stations where the UAVs take/drop off. Second, we should plan the routes of drones taking public vehicles as well as their flight routes. Third, we have to schedule multiple drones to work simultaneously. The objective of RSPMD is to minimize the total delivery time by optimizing two types of routes and drone

¹The terms buses and public vehicles are interchangeably used throughout this article.

scheduling strategies. Therefore, it is desirable to design a suitable routing and scheduling algorithm for RSPMD.

To tackle this problem, we contribute a novel PTS-assisted multidrone parcel delivery algorithm (PDD), which consists of three steps: first, customers are divided into clusters; second, suitable stations are selected and UAV flight routes are planned; and third, the global routes are constructed and optimized by combining parts of the existing routes in iterations. We also optimize the waiting time spent by the drones, which is the extra time cost incurred by the departure intervals in the bus schedules. To the best of our knowledge, we are the first to solve RSPMD. The benefits of PDD are summarized as follows. First, PDD leverages buses to expand the service range of drones, while no extra operating/fuel costs are introduced. Second, PDD utilizes multiple drones riding on buses to collaborate in parcel delivery, thereby significantly boosting efficiency and scalability. Third, we design a set of procedures for PDD, which jointly optimize the distance and time costs of drones by iteratively combining parts of the existing routes. Fourth, considering the impact of periodic bus schedules, the algorithm elegantly schedules UAVs to minimize the total time cost.

The remainder of this article is structured as follows. Section II discusses related work on VRP and the routing problem for PTS–drone collaboration. Section III formulates the PTS-assisted UAV routing problem. The algorithm to solve this problem is illustrated in detail in Section IV. Section V presents simulated experiments on the algorithm, with conclusions following in Section VI.

II. RELATED WORK

Many studies on civil applications of UAVs have been reported in the literature. Li et al. [20] elaborated on various 5G techniques based on UAVs, including network structure, joint communication, computing, and caching. Motlagh et al. [7] reviewed the applications of low-altitude UAVs in IoTs, focusing on the data collection, processing, and communication capabilities of UAV networks. Otto et al. [21] highlighted drone applications in industry, agriculture, and transportation, as well as the key technologies to be realized. As can be seen from [22], [23], [24], and [25], the various applications, especially parcel delivery, are very promising.

In recent years, the UAV-based parcel delivery approach has received a lot of attention, and researchers have studied the routing problems associated with this approach. Dorling et al. [26] established a multi-UAV logistics delivery model considering drone reuse payload and battery energy. Khoufi et al. [27] summarized the routing problems of various truck–drone collaborations in parcel delivery. The work [10], [28], [29] transported drones by trucks to expand the scope of parcel delivery. Our previous work [30] proposed *flight segment* to optimize the truck–drone collaboration, which can effectively reduce the flight distance and delivery time of truck-carried UAVs. However, all the above studies utilize trucks to assist drones, which is very different from the PTS we employ.

Only little work on the routing problem for PTS–drone collaboration was reported. The work [31] combined drone-delivery with autonomous mobility, such as taxis, crowd-sourced transportations, and car-sharing platforms, to solve the last-mile delivery problem. Huang et al. [14] presented a time-dependent scheduling problem for a vehicle–drone collaborative system considering the time for recharging and developed a dynamic programming-based algorithm to find the optimal solution. Trotta et al. [32] modeled the UAV scheduler as a mixed-integer linear programming formula and solved the routing problem for UAV in a greedy manner. Studies [14], [15], [16], [17] only discussed the situation of a single drone riding public transportation to deliver packages, lacking the research on multiple drones working simultaneously. Choudhury et al. [18] established a multidrone delivery model, but without considering the constraints of drone capacity and payload, which are very important parameters for drone delivery.

The routing and scheduling problem for PTS-assisted multidrone delivery is similar to a VRPMT, which is a variation of VRP. The work [33] presented a probabilistic technique to optimize the local search that can be applied to solve the VRP. Based on the adaptive memory procedure (AMP) [33], Olivera and Viera [34] proposed an improved algorithm to solve the VRPMT. Inspired by [33] and [34], Hu et al. [1] obtained good solutions for UAVs by combining routes and anchor points selected from the memory and assigned them to UAVs. Cheikh et al. [35] proposed a variable neighborhood search algorithm that utilizes four neighborhood structures to minimize the length of the routes and the overtime. Different from the previous work, the PTSs we employ have fixed routes and schedules. Therefore, the algorithms to solve the traditional VRP is not directly applicable to our problem.

Some work focused on the impact of PTS schedules and routes, although the majority of them did not use drones. Fu and Rilett [36] and Li et al. [37] carried out detailed research on the actual routes, traffic conditions, and station redundancy. In [14] and [38], the parcels or drones ride the bus according to the established timetables. Inspired by these studies, in this article, we formulate the problem as a VRPMT considering fixed bus schedules. PDD not only employs an improved AMP algorithm to construct global routes but also reduces time and distance costs by optimizing the waiting time spent by drones.

III. PROBLEM FORMULATION

In our scenario, all parcels are stored in a depot and will be delivered by UAVs to customers distributed in a large region. Each customer must be served once by a UAV, which can serve multiple customers on a single trip. However, the power limitations of drones prevent them from making long-distance deliveries. We employ public vehicles (e.g., buses) to transport the UAVs to compensate for their poor endurance. The public vehicles can be trains, buses, shared cars, subways, etc. It is assumed that UAVs do not consume their power when riding a bus. As shown in Fig. 1, UAVs with multiple parcels depart from the depot and take the buses for delivery. After the drones complete the delivery, they return to the depot by

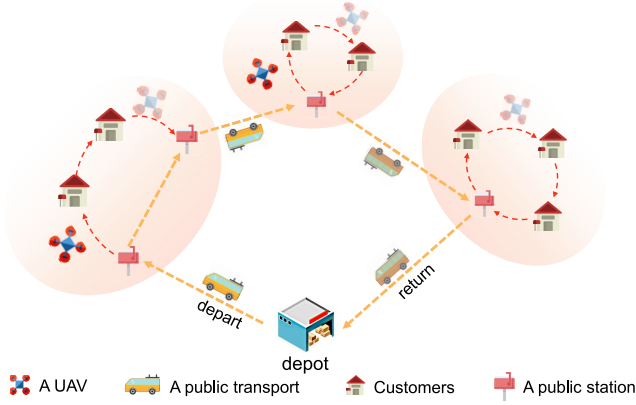


Fig. 1. Illustration of PTS-assisted multidrone parcel delivery.

buses and prepare for new tasks. It is worth noting that the public vehicles have fixed schedules, which include the departure intervals; therefore, UAVs sometimes need to wait at the stations. After the deliveries are completed, all drones will return to the depot. For simplicity, we make the following assumptions.

- 1) The coordinates of the customers and the weights of the parcels are known in prior.
- 2) Both buses and UAVs travel at constant speeds.
- 3) There are ample stations for drone interchanges and their coordinates are known.
- 4) The departure intervals at each station are known and the same.
- 5) Roads and stations form a connected graph.
- 6) The time spent by drones serving customers, replacing batteries, boarding/dropping the vehicle is negligible.
- 7) UAVs can board buses only when they stop at the stations; otherwise, UAVs must wait at the stations.

The notation and terminology used throughout this article are explained in Table I. Suppose $S = \{1, 2, \dots, c\}$ denotes the customer set. We use both nodes 0 and $c + 1$ to represent the depot, implying that the UAV needs to depart from and return to the depot. Let $S_{all} = \{0, 1, 2, \dots, c + 1\}$ denote the set of all nodes that must be visited in the area. The set $M = \{m_1, m_2, \dots, m_{n_p}\}$ denotes the weight of each customer's parcel. Let set $P = \{1, 2, \dots, s\}$ denote the public transportation stations. Notice that the UAVs can only board/drop off at the station locations. The distance between nodes n_p, n_q is expressed as d_{n_p, n_q} . Similarly, the distance between the customer n_p and the station P_j is denoted as d_{n_p, P_j} , and the distance between two stations is indicated as d_{P_i, P_j} . In addition, the parameter V_T signifies the speed of public transportation. The maximum load, the longest flight distance, and the flight speed of UAV are, respectively, denoted as M_{max} , D_{max} , and V_u . In this article, the departure intervals in the bus schedules at each station are the same and they are expressed as t_p . If the vehicle enters the station at moments $a + kt_p$ ($k = 0, 1, \dots$) and leaves the station at moments $b + kt_p$, the periods when it stops at each station are denoted as $[a + kt_p, b + kt_p]$, which means that the UAV must arrive at the station within $[a + kt_p, b + kt_p]$, $k = 0, 1, \dots$, to take the bus. In general,

TABLE I
NOTATION AND TERMINOLOGY

Notation	Definition
S	a set of customers to be served
S_{all}	a set of customers and the depot
M	a set of customers' parcel weight
P	a set of public transportation station
d_{n_p, n_q}	the distance between nodes n_p, n_q
d_{n_p, P_j}	the distance between the customer n_p and the station P_j
d_{P_i, P_j}	the distance between stations P_i, P_j
V_T	the speed of the public vehicle
M_{max}	the maximum payload of the UAV
D_{max}	the hovering distance of the UAV
V_u	the speed of the UAV
t_p	the departure intervals of public transportation
$[a + kt_p, b + kt_p]$	the period the public transport stops at the station
clu	the result of clustering customers
R_{clu}	a set of UAVs' flight routes in clusters
L_{clu}	a set of UAVs' route length in the clusters
$T_{clu}(i)$	the time spent by UAV flying in the cluster i
$y_{vh}(n_p, n_q)$	binary indicating whether n_p is right next to n_q
$y_{vh}(n_p, P_j)$	binary indicating whether n_p is right next to P_j
$y_{th}(P_i, P_j)$	binary indicating whether P_i is right next to P_j
R_{gl}	a set of global routes
$T_i(R_{gl}^i)$	the time spent by UAV on public transportation in R_{gl}^i
$T_w(R_{gl}^i)$	the total time the UAV waits at the stations in R_{gl}^i
T	the overall objective function
x_{clu_j, R_{gl}^i}	binary indicating whether clu_j is visited in R_{gl}^i
x_{n_p}	binary indicating whether n_p is visited

the objective of this article aims to minimize the total time for UAV parcel delivery by jointly optimizing UAV flight route, station selection, global routes, and UAV task assignment. Based on the above scenario descriptions, we make a detailed analysis of RSPMD and simplify the problem into three parts.

A. Division of Flight Area and Station Selection

Customers are not evenly distributed in the area. For example, residential areas with a lot of parcels may require multiple drones for delivery, compared to areas with low density. Therefore, it is necessary to partition the flight area according to the payload capacities of the UAVs. The flight area of UAVs in one trip is limited by the maximum load M_{max} and the longest flight distance D_{max} . When dividing the areas, the parcel weights and the customer locations need to be considered. By splitting the flight area, the customers will be divided into clusters. The customers in each cluster are denoted as $c_n = \{1, 2, \dots, c\}$. The set of clusters is indicated as $clu = \{c_1, c_2, \dots, c_k\}$.

After division, it is necessary to select appropriate stations for UAVs to board/drop off. Appropriate stations are convenient for UAVs to visit/leave clusters, which can reduce the energy consumption of UAVs in flight. The stations P_i and P_j will be selected from the candidate stations as the UAV drop-off station and the UAV pick-up station, respectively. In other words, P_i and P_j are the starting and ending points of the UAV flight route that must be visited. The selected stations are added to a cluster c_n , which is denoted as $c_n = \{P_i, 1, 2, \dots, c, P_j\}$.

B. Cost in the UAV Flight Routes

After determining the flight area, the order of unvisited points within it needs to be determined, which is a routing problem for the UAVs. The set of UAV flight route is denoted as $R_{clu} = \{R_{clu}^1, R_{clu}^2, \dots, R_{clu}^k\}$, where R_{clu}^1 represents the flight routes in cluster c_1 . The parameter L_{clu}^i denotes the length of route R_{clu}^i . Besides, the auxiliary decision variable $y_{vh}(n_p, n_q) \in \{0, 1\}$ denotes whether customer n_p is right next to customer n_q . Let $y_{vh}(n_p, P_j) \in \{0, 1\}$ signifies whether the station P_j is right next to the customer n_p . The length of routes and the flight time in each cluster can be expressed as

$$L_{clu}^i = \sum_{n_p \in c_i} \sum_{n_q \in c_i} y_{vh}(n_p, n_q) d_{n_p, n_q} + \sum_{n_p \in c_i} \sum_{P_j \in P} y_{vh}(n_p, P_j) d_{n_p, P_j} \quad (1)$$

$$T_{clu}(i) = \frac{L_{clu}^i}{V_u}. \quad (2)$$

It is obvious that the subproblem is an *open-loop* TSP since the starting and ending points of the flight route can be different stations. With the objective of shortening the flight distance of UAVs, the routing problem for the UAVs can be expressed as follows:

$$\min L_{clu}^i = \min \left\{ \sum_{n_m \in c_i} \sum_{n_k \in c_i} y_{vh}(n_m, n_k) d_{n_m, n_k} + \sum_{n_p \in c_i} \sum_{P_j \in P} y_{vh}(n_p, P_j) d_{n_p, P_j} \right\}. \quad (3)$$

C. Cost in the Global Routes

The global routes include the UAV flight routes and the routes of the UAVs on the buses. Considering each cluster as an unvisited point, the planning of the global routes is equivalent to determining the order in which the UAV visits each cluster. This subproblem is similar to a VRPMT, except that the impact of the departure intervals on the solution time cost should be considered. By solving it, we can obtain n feasible routes for UAV to visit the clusters. The set of routes is denoted as $R_{gl} = \{R_{gl}^1, R_{gl}^2, \dots, R_{gl}^n\}$, where R_{gl}^1 represents the first route of UAV. Notice that each route is composed of the depot, necessary stations, and customer clusters. Hence, the route R_{gl}^i can be indicated as $R_{gl}^i = \{0, clu_i, P_i, \dots, P_j, c+1\}$. The time spent by the UAV on the buses in the route R_{gl}^i is given as follows:

$$T_t(R_{gl}^i) = \frac{\sum_{P_p \in R_{gl}^i} \sum_{P_q \in R_{gl}^i} y_{th}(P_p, P_q) d_{P_p, P_q}}{V_u} \quad (4)$$

where $y_{th}(P_p, P_q) \in \{0, 1\}$ equals 1 if the station P_p is right next to the station P_q .

In the route R_{gl}^i , the time spent by the drone waiting for the buses at each station is denoted as $T_w(R_{gl}^i)$. Knowing the departure intervals in the bus schedules, the waiting time can be calculated.

Based on (2) and (4), the total time for parcel delivery can be formulated as follows:

$$T = \min \sum_{R_{gl}^i \in R_{gl}} \left[T_t(R_{gl}^i) + T_w(R_{gl}^i) + \sum_{c_j \in R_{gl}^i} T_{clu}(j) \right] \quad (5)$$

s.t.

$$\sum_{n_p \in c_i} m_{n_p} \leq M_{\max} \quad (6)$$

$$d_{n_p, n_q} \leq \frac{D_{\max}}{2} \quad (7)$$

$$\sum_{R_{gl}^i \in R_{gl}} x_{clu_j, R_{gl}^i} = 1 \quad \forall clu_j \in clu \quad (8)$$

$$\sum_{P_j \in P} x_{n_p, P_j} = 1 \quad \forall n_p \in S \quad (9)$$

$$\sum_{c_i \in clu} \sum_{n_p \in c_i} x_{n_p} = c. \quad (10)$$

Equation (6) ensures that the total weights of the parcels in each cluster are no greater than the maximum load of the UAV. Equation (7) emphasizes that UAV can directly reach customers in clusters. In (8), the term $x_{clu_j, R_{gl}^i} \in \{0, 1\}$ indicates whether clu_j is visited in R_{gl}^i , and it ensures that each customer cluster belongs to only one global route. Equation (9) ensures that each customer node only matches one station. In (10), the auxiliary decision variable $x_{n_p} \in \{0, 1\}$ equals 1 if the customer is visited by UAV. Equation (10) ensures that every customer is visited.

IV. ALGORITHM

The routing and scheduling problem of RSPMD that we study is similar to VRP. Compared to truck-assisted, it is a more special two-layer VRPMT problem because public transportation has fixed routes and schedules. Therefore, based on the idea of divide and conquer, we use clustering and path planning to construct local solutions in the first layer. Then, we further combine it into a global solution through collaboration and scheduling of multiple UAVs. The scheduling of multiple UAVs is a bin-packing problem (BPP), where each UAV may be assigned multiple timing tasks in order to improve system efficiency. In this section, we propose a novel routing and scheduling algorithm, referred to as the PTS-assisted multidrone parcel delivery (PDD) algorithm. As demonstrated in Section III, our problem is NP-hard and needs to be solved by a specialized and efficient heuristic. Fig. 2 is the flowchart of PDD, which contains three steps: 1) customers clustering; 2) UAV flight route construction; and 3) global route construction. First, the customers are divided into clusters according to the parcel weights and the customer locations. Second, the algorithm matches the stations to each cluster and plans the drone flight routes. Third, it determines the order of the clusters and plans the routes for the drones to take public vehicles. In this step, we obtain improved solutions by inserting and removing partial solutions. The pseudocodes of PDD are shown in Algorithm 1.

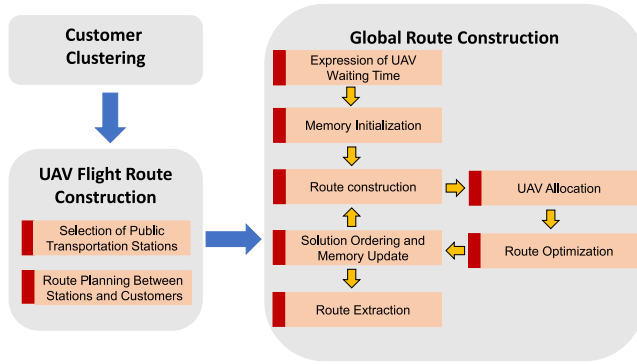


Fig. 2. Flowchart of the PDD algorithm.

Algorithm 1: PTS-Assisted Multidrone Parcel Delivery

```

1 Input:  $S = \{1, 2, \dots, c\}$ ;
2 Output:  $r_{final}$ ;
3  $clu \leftarrow \text{call } customer\_clustering(S)$ ;
4  $R_{clu} \leftarrow \text{call } obtain\_intra\_cluster\_routes(clu, P)$ ;
5  $M \leftarrow \text{call } initialization\_memory(R_{clu})$ ;
6 for  $i = 1 \rightarrow GL^{iter}$  do
7    $U_{uav} \leftarrow \text{call } allocate\_UAV(M)$ ;
8    $r_{opti} \leftarrow \text{call } routes\_optimization(U_{uav})$ ;
9    $M \leftarrow M \cup \{r_{opti}\}$ ;
10  sort  $M$  according to  $v(R_{gl}^i)$ ;
11  $r_{final} \leftarrow \text{call } routes\_extraction(M)$ ;
12 return  $r_{final}$ ;

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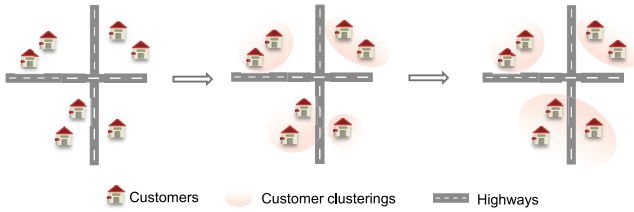


Fig. 3. Clustering process.

A. Customer Clustering

Given the limited flight distance and load capacity of the UAV, customers need to be partitioned such that UAVs can deliver more parcels in one flight. Based on the improved AGNES [39] algorithm, all customers in the area are divided into clusters. Fig. 3 shows the clustering process while Algorithm 2 presents its pseudocodes. Each customer is first regarded as an isolated cluster and then combined into a larger cluster according to their locations and the total weights of parcels. Therefore, there is no “isolated” customer after clustering. Through clustering, the smallest unit for UAV delivery is transformed from a single customer into a cluster containing multiple customers.

The algorithm prioritizes the fusion of the two clusters with the smallest average distance. The average distance $d_{avg}(c_i, c_j)$ [39] between the two clusters is defined as

Algorithm 2: customer_clustering(S)

```

1 begin initialize set of the clusters  $clu = \{c_1, c_2, \dots, c_k\}$ ,
   mean distance matrix  $D$ ;
2 for  $s_i \in S$  do
3    $c_i \leftarrow s_i$ ;
4 update  $D$  by  $d_{avg}(c_i, c_j) = \frac{1}{|N_{c_i}| |N_{c_j}|} \sum_{n_p \in c_i} \sum_{n_q \in c_j} d_{n_p, n_q}$ ;
5  $q \leftarrow \text{size of } clu$ ;
6 while  $q > k$  do
7   obtain  $c_i, c_j$  with the minimum  $d_{avg}(c_i, c_j)$ ;
8   if  $\sum_{n_p \in c_i, c_j} m_{n_p} < M_{max}$  then
9      $c_i \leftarrow c_i \cup c_j$ ;
10    renumber  $clu$ ;
11    update  $D$  according to  $d_{avg}(c_i, c_j)$ ;
12     $q \leftarrow q - 1$ ;
13  else
14     $d_{avg}(c_i, c_j) \leftarrow \infty$ ;
15 return  $clu$ ;

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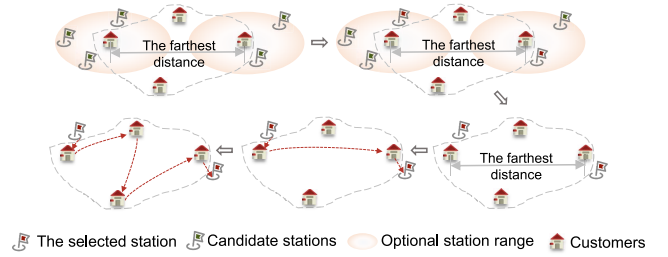


Fig. 4. UAV subroute construction.

$$d_{avg}(c_i, c_j) = \frac{1}{|N_{c_i}| |N_{c_j}|} \sum_{n_p \in c_i} \sum_{n_q \in c_j} d_{n_p, n_q} \quad (11)$$

where the parameter N_{c_i} denotes the number of customers in the cluster c_i . If the total weights of the parcels for both clusters are less than M_{max} , the two clusters are merged into one; otherwise, they are not merged. Repeat the above fusion process until the number of clusters is reduced to the target number k .

B. UAV Flight Route Construction

The construction of the flight routes in clusters mainly contains two steps (as shown in Algorithm 3): 1) selecting appropriate stations for each cluster and 2) designing the flight routes in each cluster. The process is shown in Fig. 4.

1) *Selection of Public Transportation Stations:* Appropriate stations can reduce the detour routes of UAVs. The algorithm first finds the two customers with the longest distance d_{n_p, n_q} in the cluster to ensure that the other customers are distributed in a circle with the longest d_{n_p, n_q} as the diameter. Then, it matches the two customers to the nearby stations, respectively. When customer n_p matches the stations, a circle is formed with n_p as the center and ϵD_{max} as the radius, where $0 < \epsilon < 1$. The parameter ϵ is used to adjust the size of the circle. All stations in this circle are treated as candidate stations for n_p and are denoted as

Algorithm 3: obtain_intra-cluster_routes(clu, P)

```

1 begin initialize  $R_{clu}, \epsilon, P_{alter}$ ;
2 for  $c_k \in clu$  do
3   select  $n_p, n_q$  with the maximum  $d_{n_p, n_q}$  from  $c_k$ ;
4   form a circle with  $n_p$  as center and  $\epsilon D_{max}$  as radius;
5   form a circle with  $n_q$  as center and  $\epsilon D_{max}$  as radius;
6   for  $P_i \in P$  do
7     if  $d_{n_p, P_i} < \epsilon D_{max}$  then
8        $P_a^{n_p} \leftarrow P_a^{n_p} \cup \{P_i\}$ 
9     if  $d_{n_q, P_i} < \epsilon D_{max}$  then
10       $P_a^{n_q} \leftarrow P_a^{n_q} \cup \{P_i\}$ 
11   select  $P_i$  with the minimum  $d_{n_p, P_i}$  from  $P_a^{n_p}$ ;
12   select  $P_j$  with the minimum  $d_{n_q, P_j}$  from  $P_a^{n_q}$ ;
13    $R_{clu}^k \leftarrow R_{clu}^k \cup \{P_i, P_j\}$ ;
14   update  $L_{clu}^k$ ;
15   for  $n_t \in c_k$  do
16      $R_{clu}^k \leftarrow R_{clu}^k \cup \{n_t\}$  such that  $L_{clu}^k$  is minimal;
17     update  $L_{clu}^k$ ;
18 return  $R_{clu} = \{R_{clu}^1, R_{clu}^2, \dots, R_{clu}^k\}$ ;

```

$P_{alter} = \{P_a^1, P_a^2, \dots, P_a^{n_p}\}$. Among them, the station with the minimum d_{n_p, P_j} is selected as the starting or ending point of UAV flight routes. The remaining candidate stations are stored in memory and participate in global route optimization (lines 3–13 of Algorithm 3). It is worth noting that when two nodes are close, they are probably matched to the same station.

2) *Route Planning Between Stations and Customers:* This step plans the UAV flight routes in clusters. As presented in Section III-B, the routing problem for the UAVs is an *open-loop* TSP, which is solved by an improved insertion algorithm. Suppose the matched stations are P_i and P_j . The planning of the flight routes consists of the following four steps.

- 1) The stations P_i and P_j serve as the starting and ending points of the route, respectively (P_i is the starting point and P_j is the ending point or vice versa). Let R_{clu}^i denote routes for UAV to serve the customers in cluster c_i .
- 2) Add unvisited customers to the route, making the total length L_{clu}^i of the current route the shortest.
- 3) Update the routes R_{clu}^i, L_{clu}^i and the set of unvisited customers.
- 4) If the set of unvisited customers is nonempty, repeat 2); otherwise, the UAV flight route construction ends.

C. Expression of UAV Waiting Time

Different from private-owned trucks, buses have fixed schedules and travel routes. Therefore, it is necessary to calculate not only the distance cost of the global routes but also the waiting time spent by drones at stations. The algorithm converts the waiting time of UAVs into a function that can be optimized in subsequent algorithms.

The time elapses of the stages in the route are shown in Fig. 5. In the figure, columns of the same colors indicate that

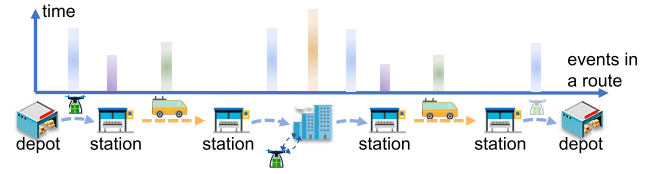


Fig. 5. Illustration of time period of a global route.

the drones are in the same stages. The different lengths of the purple columns mean that the drones have different waiting times at the stations, which is caused by the departure intervals in bus schedules. Therefore, it is not appropriate to use the time average to represent the waiting time. Ignoring the waiting time of the drones, this problem is similar to a traditional VRPMT. By solving it, n global routes will be obtained, indicated as $R_{gl} = \{R_{gl}^1, R_{gl}^2, \dots, R_{gl}^n\}$. As assumed in Section III, each station is available for UAVs during the time period $[a + kt_p, b + kt_p]$. To calculate the waiting time of UAVs at the stations, the parameter $\tau_a(P_i, R_{gl}^i)$ denotes the moment when UAV arrives at the station P_i in the route R_{gl}^i , and $\tau_d(P_i, R_{gl}^i)$ denotes the moment UAV leaves P_i , the period $t(P_i, R_{gl}^i)$ that UAV waits at the P_i station in the route R_{gl}^i is

$$t(P_i, R_{gl}^i) = \tau_d(P_i, R_{gl}^i) - \tau_a(P_i, R_{gl}^i). \quad (12)$$

Built on (12), the waiting time of the drone in the route R_{gl}^i can be expressed as

$$\begin{aligned} T_{R_{gl}^i}^P &= \sum_{P_i \in P} t(P_i, R_{gl}^i) y_{R_{gl}^i, P_i} \\ &= \sum_{P_i \in P} [\tau_d(P_i, R_{gl}^i) - \tau_a(P_i, R_{gl}^i)] y_{R_{gl}^i, P_i} \end{aligned} \quad (13)$$

where $y_{R_{gl}^i, P_i} \in \{0, 1\}$ equals one if UAV stops at the P_i in R_{gl}^i . The waiting time $T_{R_{gl}^i}^P$ will be used as one of the metrics to evaluate the performance of each solution and participate in the construction and optimization of the global routes. It is worth noting that the waiting time is affected by many factors, including the time spent by UAV serving customers, station selection, and the departure intervals of the PTS. According to the above-mentioned factors, the recursive formula of the arrival and departure time of UAV can be obtained. There are two situations for UAVs to reach the station: 1) arrive at this station after leaving the previous one and 2) arrive at this station after serving the last customer. Then, the expressions of $\tau_a(P_i, R_{gl}^i)$ in two cases are

$$\tau_a(P_i, R_{gl}^i) = \tau_d(P_{i-1}, R_{gl}^i) + \frac{d_{P_i, P_{i-1}}}{V_T} \quad (14)$$

$$\tau_a(P_i, R_{gl}^i) = \tau_d(P_{i-1}, R_{gl}^i) + \sum_{i \in k} T_{clu}^i x_{clu_i, R_{gl}^i}. \quad (15)$$

In the first case, $\tau_a(P_i, R_{gl}^i)$ can be obtained by calculating the time spent by drones on public vehicles. In the second case, UAV reaches the station after the parcel delivery is completed. Therefore, it is necessary to add the time spent by the drone flying in the clusters.

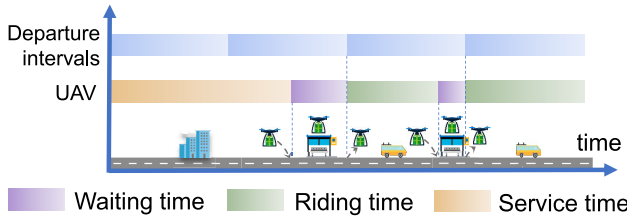


Fig. 6. Illustration of time UAV waits at a station.

In the same way, the process of UAV leaving the station is also classified into two situations: 1) depart from the previous station to reach this station then leave and 2) after serving the customer, arrive at this station then leave. The expressions are as follows:

$$\tau_d(P_i, R_{gl}^i) = \tau_d(P_{i-1}, R_{gl}^i) + \left\lceil \frac{dp_{i,P_{i-1}}}{V_T t_p} \right\rceil t_p \quad (16)$$

$$\tau_d(P_i, R_{gl}^i) = \tau_d(P_{i-1}, R_{gl}^i) + \left\lceil \frac{\sum_{i \in k} T_{clu}^i x_{clu_i, R_{gl}^i}}{t_p} \right\rceil t_p. \quad (17)$$

Although the UAV can reach the station at any time, it can only leave the station at the time when the bus departs, i.e., the UAV can only leave the station at the time of $b + kt_p$. As shown in Fig. 6, the blue rectangles denote fixed departure intervals. Regardless of when the drone arrives at the station, it can only exit the station when the vehicle leaves.

D. Global Route Construction

In this section, the global routes are planned considering the PTS schedules and subsequently optimized to obtain good solutions. Simplifying this problem to a VRPMT, we use an improved AMP algorithm to obtain the global routes, which consists of five steps: 1) memory initialization; 2) UAV allocation; 3) routes optimization; 4) solution ordering and memory update; and 5) routes extraction. After the previous construction, the stations in global route R_{gl}^i can be classified into two categories based on whether they match customers. The stations that are not directly connected to customers are referred to as transfer stations. In the final solution, there are multiple scenarios that can occur at a given moment for drones. Some drones are delivering packages in a cluster, some are waiting for a ride at a station, and others are on the buses.

1) *Memory Initialization*: An improved sweep algorithm is used to construct feasible global routes as the initial memory. We first group all clusters to ensure that the parcel weights in each route are less than $M_{\max}$. Then, the order of the clusters within each group is determined. The memory initialization process is shown in Fig. 7.

We need to group customer clusters. The clusters are indicated by the matched stations in Section IV-B. To ensure the randomness of the solutions, the algorithm starts grouping from any cluster. If the parcel weights in each route are less than $M_{\max}$, add a new cluster to the current route; otherwise, assign the cluster to a new route. It is worth noting that other stations cannot be inserted between two stations matching the same cluster. In a single route, the current problem is simplified to a classic TSP. To tackle it, we use the 2-opt algorithm

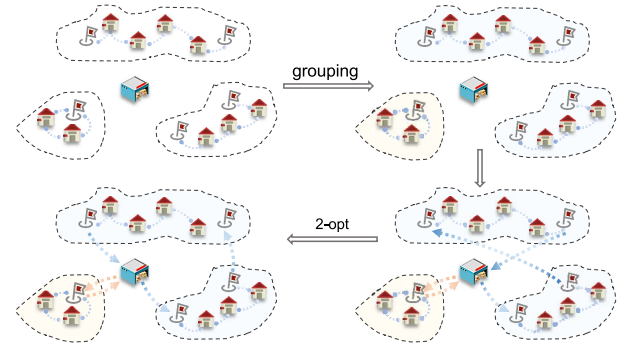


Fig. 7. Initial route construction.

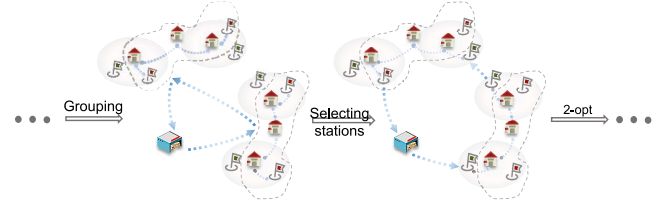


Fig. 8. Differences in constructing routes using candidate stations.

to adjust the order of the stations. It is obvious that when the distance between the two stations is greater than $D_{\max}$, UAV needs to take public vehicles.

Let the candidate stations retained in Section IV-B participate in the global route construction to enrich the memory with viable solutions. The steps for cluster grouping and 2-opt algorithm optimization are the same as the construction process in Fig. 7. The difference from the above construction is shown in Fig. 8. After the clusters are grouped, candidate stations are randomly added to the route, and reordered by the 2-opt. In this article, we refer to the memory initialization step as *initialization_memory()*.

2) *UAV Allocation*: The initial routes need to be assigned to drones, which is a BPP. In the scheduling process of multiple UAVs, an improved best fit reduction (BFD) algorithm is employed to tackle this problem to achieve better UAV collaboration as possible. The allocation process is explained in Algorithm 4. According to customer demand and the maximum load of UAVs, the minimum number of UAVs can be calculated as

$$N_{\min} = \left\lceil \frac{\sum_{n_p \in S} m_{n_p}}{M_{\max}} \right\rceil. \quad (18)$$

In each allocation, In order to increase the utilization of UAVs, a new task will be assigned after the UAV completes its current task. The route that takes longer will be assigned to the drone with the shorter completion time to obtain the shortest total average time (lines 5–10 of Algorithm 4). If the current number of UAVs is insufficient, increase idle drones until all routes can be allocated (lines 11–15 of Algorithm 4).

3) *Route Optimization*: The above steps do not consider the bus schedules, and the performance of the global routes is not outstanding. Therefore, an improved tabu search algorithm is used to optimize the solutions. We evaluate the pros and cons of each route by the weighted sum of UAV flight distance

Algorithm 4: allocate_UAV(M)

```

1  $n \leftarrow N_{\min}$  by  $N_{\min} = \lceil \frac{\sum_{np \in S} m_{np}}{M_{\max}} \rceil$ ;
2 begin initialize  $U_{uav} = \{u_1, u_2, \dots, u_n\}$ ;
3 sort  $R_{gl}^i \in M$  in descending order according to
    $T_t(R_{gl}^i) + T_w(R_{gl}^i) + \sum_{c_j \in R_{gl}^i} T_{clu}(j)$ ;
4 for  $R_{gl}^i \in M$  do
5   sort  $U_{uav}$  in nondescending order according to  $t_{uav}^n$ ;
6   for  $u_k \in U_{uav}$  do
7     if  $\sum_{np \in R_{gl}^i} m_{np} + \sum_{R_{gl}^i \in u_k} m_{np} \leq M_{\max}$  then
8        $u_k \leftarrow u_k \cup \{R_{gl}^i\}$ ;
9        $t_{uav}^k \leftarrow$ 
10         $t_{uav}^k + [T_t(R_{gl}^i) + T_w(R_{gl}^i) + \sum_{c_j \in R_{gl}^i} T_{clu}(j)]$ ;
11     break;
12   if  $R_{gl}^i \notin U_{uav}$  then
13      $n \leftarrow n + 1$ ;
14      $u_n \leftarrow u_n \cup \{R_{gl}^i\}$ ;
15      $t_{uav}^n \leftarrow t_{uav}^n + [T_t(R_{gl}^i) + T_w(R_{gl}^i) + \sum_{c_j \in R_{gl}^i} T_{clu}(j)]$ ;
16      $U_{uav} \leftarrow U_{uav} \cup \{u_n\}$ 
17 return  $U_{uav}$ ;

```

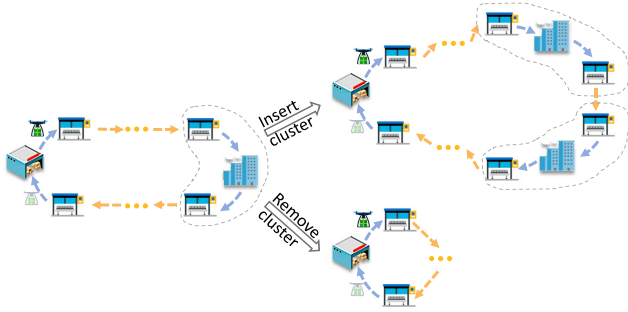


Fig. 9. Customer clusters and connected stations optimization.

in (2), mission time in (5), and UAV waiting time in (13). The evaluation function is as follows:

$$v(R_{gl}^i) = T + \alpha \sum_{c_k \in clu} L_{clu}^i + \beta T_{R_{gl}^i}^P \quad (19)$$

where α is the penalty factor corresponding to the UAV flight distance, and β is the penalty factor corresponding to the UAV waiting time. The nodes in each global route can be classified into two categories: 1) the clusters and the stations connected to them and 2) the transfer stations. We will optimize the two types of stations separately, which are shown in Algorithm 5. The following operations will be performed in routes $R_{gl}^i = \{s, \dots, P_i, c_i, P_{i+1}, \dots, s\}$, and $R_{gl}^j = \{s, \dots, P_j, c_j, P_{j+1}, \dots, s\}$.

- 1) *Optimizing Customer Clusters and Connected Stations:* *Insert_customer_cluster* and *Remove_customer_cluster* are used to optimize existing routes, which are shown in Fig. 9. We define the radius r_{c_i} of the search centered

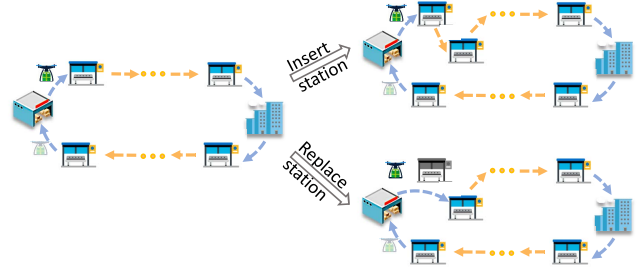


Fig. 10. Transfer stations optimization.

on the cluster c_i as

$$r_{c_i} = \delta \max_{np, n_q \in c_i} d_{np, n_q}. \quad (20)$$

Insert_Customer_Cluster: When c_i is the search center, the operation searches within the radius r_{c_i} . If c_j is searched and its parcel weights are less than $M_{\max} - \sum_{np \in R_{gl}^i} m_{np}$, c_j and its connected two stations P_j, P_{j+1} can be inserted after P_{i+1} in the route R_{gl}^i .

Remove_Customer_Cluster: For route R_{gl}^i , c_j and its connected two stations P_j, P_{j+1} are deleted from route R_{gl}^i at the same time. The remaining stations and customer clusters are visited in the original order.

- 2) *Optimizing Transfer Stations:* As presented in Fig. 10, we use *Insert_station* and *Remove_station* to optimize the routes. The search radius r_{p_i} of the station P_i is defined as

$$r_{p_i} = \epsilon D_{\max}. \quad (21)$$

Insert_Station: Insert the station P_j in the search range of P_i after P_i .

Remove_Station: P_i is deleted from R_{gl}^i . The remaining stations and clusters are visited in the original order.

In one iteration, lines 5–17 of Algorithm 5 perform the above operations on the route R_{gl}^i to generate its neighborhood N_{neigh} . Next, route R_{gl}^{i*} is selected from N_{neigh} and compared with R_{gl}^i (lines 18 of Algorithm 5). If R_{gl}^{i*} takes less time and distance cost, it is added to r_{opti} and the tabu list is updated at the same time (lines 19–24 of Algorithm 5).

4) *Solution Ordering and Memory Update:* The optimized routes are added to the memory and sorted according to $v(R_{gl}^i)$ in (19) such that the optimal solution appears on the top of the memory as much as possible. Due to the limited payload of UAVs, the parcel weight in each route is the first criterion for ranking. If $\sum_{np \in R_{gl}^i} m_{np}$ exceeds the maximum load of UAV, the route will be regarded as an infeasible solution. Second, routes are sorted by function value $v(R_{gl}^i)$. The route with a smaller $v(R_{gl}^i)$ will be ranked higher in the memory.

5) *Route Extraction:* The final solution is constructed by selecting routes from the memory with a certain probability. In the previous steps, superior solutions are ranked at high positions in the memory and have high probabilities of being extracted. The procedure to extract the optimal routes is shown in Algorithm 6.

Algorithm 5: routes_optimization(U_{uav})

```

1 begin initialize  $List_{tabu}, N_{neigh}$ ;
2  $U_{best} \leftarrow U_{uav}$ ;
3  $r_{opti} \leftarrow U_{uav}$ ;
4 for  $i = 1 \rightarrow TS_{iter}$  do
5   select  $R_{gl}^i$  with the maximum  $v(R_{gl}^i)$  from  $U_{best}$ ;
6   for  $c_k \in R_{gl}^i$  do
7     if  $\{\sum_{n_p \in c_j} m_{n_p} < M_{max} - \sum_{n_p \in R_{gl}^i} m_{n_p}\} \& \{d_{avg}(c_k, c_j) < r_{c_k}\}$  then
8        $R_{gl}^{i*} \leftarrow$  apply Insert_customer_cluster
        operation on  $R_{gl}^i$ ;
9     else
10       $R_{gl}^{i*} \leftarrow$  apply Remove_customer_cluster
        operation on  $R_{gl}^i$ ;
11     $N_{neigh} \leftarrow N_{neigh} \cup \{R_{gl}^{i*}\}$ ;
12  for  $P_k \in R_{gl}^i$  do
13    if  $d_{P_i, P_j} < r_{P_i}$  then
14       $R_{gl}^{i*} \leftarrow$  apply Insert_station operation on  $R_{gl}^i$ ;
15    else
16       $R_{gl}^{i*} \leftarrow$  apply Remove_station operation on
         $R_{gl}^i$ ;
17     $N_{neigh} \leftarrow N_{neigh} \cup \{R_{gl}^{i*}\}$ ;
18  select  $R_{gl}^{i*}$  with the minimum  $v(R_{gl}^{i*})$  from  $N_{neigh}$ ;
19  if  $v(R_{gl}^{i*}) < v(R_{gl}^i)$  then
20     $r_{opti} \leftarrow r_{opti} \cup \{R_{gl}^{i*}\}$ ;
21     $U_{best} \leftarrow r_{opti}$ ;
22     $r_{opti} \leftarrow r_{opti} \setminus \{R_{gl}^i\}$ ;
23     $N_{neigh} \leftarrow N_{neigh} \setminus \{R_{gl}^i\}$ ;
24  update  $List_{tabu}$  according to  $N_{neigh}$ ;
25 return  $r_{opti}$ ;

```

V. PERFORMANCE EVALUATION

An evaluation study is presented in this section to assess the performance of the proposed PDD. To show the advantages of PDD, we not only conducted a self-comparison of the algorithm but also selected two other basic algorithms for comparison: 1) K -means and GA (KGA) [28] and 2) a relaxed version of AMP (relaxed-AMP) [34]. KGA jointly solves a two-echelon vehicle routing problems (2E-VRPs) [40] where K -means finds the launch points of drones and GA solves the TSP for a truck. AMP is specialized in solving VRPMT. Only by lifting the limit on flying distance, an effective solution for UAV delivery routes can be obtained through AMP.

The evaluation metrics include time cost and distance cost. Among them, the time cost is calculated from the UAVs leaving the depot, completing the delivery to returning to the depot. In particular, we focus on the impact of the PTS departure intervals on the mission time, i.e., the waiting time UAVs spent at the stations. Another metric distance cost refers to the distance of all routes, which includes the distance of UAV taking

Algorithm 6: routes_extraction(M)

```

1  $M' \leftarrow M, clu' \leftarrow clu$ ;
2 calculate  $P_{R_{gl}^i}$  by  $P_{R_{gl}^i} = 2^{\frac{|M'|+1-i}{|M'|(|M'|+1)}}$ ;
3 for  $R_{gl}^i \in M'$  do
4   choose  $R_{gl}^i$  according to  $P_{R_{gl}^i}$ ;
5   if  $R_{gl}^i$  is chose then
6      $r_{final} \leftarrow r_{final} \cup \{R_{gl}^i\}$ ;
7      $R_{cho} \leftarrow$  the set of the routes containing the
        clusters in  $R_{gl}^i$ ;
8      $M' \leftarrow M' \setminus R_{cho}$ ;
9     delete the clusters contained by  $R_{gl}^i$  from  $clu'$ ;
10  if  $clu' \neq \emptyset$  then
11     $R_{gl}^i \leftarrow$  solve TSP of  $clu'$  by sweep algorithm;
12     $r_{final} \leftarrow r_{final} \cup \{R_{gl}^i\}$ ;
13   $r_{final} \leftarrow$  call  $allocate\_UAV(r_{final})$ ;
14 return  $r_{final}$ ;

```

buses and the distance of UAV flying. Indubitably, the smaller the values of the two metrics, the more efficient PDD is.

In our simulation, customers are randomly distributed within a 100 km $\times$ 100 km region. The depot is located in the center of the region and stores all parcels of different weights. We set the speed of the UAV as 50 km/h, and it can fly up to 7 km under maximum payload conditions. Six drones are employed in the following simulations. The public vehicles stop at the stations for 3 min, and their travel speed is 70 km/h. In PDD, we set the tabu tenure interval range as [5, 10]. The penalty factor α and the penalty factor β are all set as 1. To study the influence of PTS on the performance of the algorithm, we either vary the number of stations among (50, 200] or the departure intervals of public vehicles between 0.2 and 0.6 h, while keeping the payload capacity of UAV at 1.5 kg and the number of customers at 300. The corresponding results are shown in Figs. 11 and 12. And we find the optimal performance of the algorithm when the number of stations is 150 and the departure interval of public vehicles is 0.4 h. In the following experiments, we either vary the number of customers among [100, 500] or the payload capacity of UAV between 1.5 and 4.5 kg, while keeping the number of stations at 150 and the departure intervals at 0.4 h. The results are shown in Figs. 13 and 14, respectively.

Fig. 11 presents the influence of the number of stations on the performance of the algorithm. From Fig. 11(b), we can see that the increase in the number of stations can significantly shorten the necessary flight distance of the drones. However, this trend flattens out when there is a higher number of stations. This means that increasing the number of stations has a decreasing effect on reducing the UAV flight distance. Fig. 11(a) and (c) shows that when the number of stations does not exceed 150, the time cost decreases by 17% and the distance cost of public vehicles decreases by 27%, while when the number exceeds 150, the time cost and the distance cost begin to increase. This is because too many stations increase

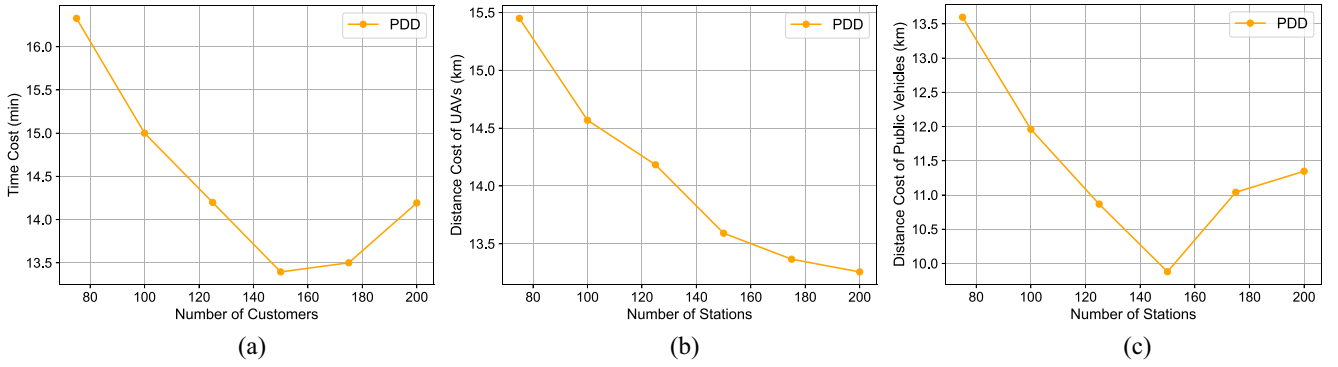


Fig. 11. Results with varying the number of stations. (a) Time cost. (b) Travel distance of UAVs. (c) Travel distance of vehicles.

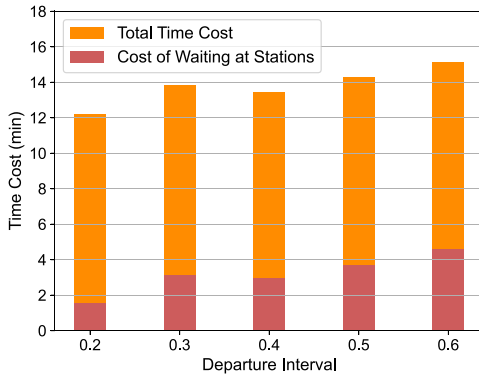


Fig. 12. Results with varying the departure intervals.

unnecessary transfers and detours of UAVs, which in turn leads to an increase in the time cost and distance cost of public vehicles. To balance model efficiency and equipment costs, the number of stations in the following experiments is set as 150.

Fig. 12 depicts the effect of the bus schedules on the performance of PDD. As the departure interval increases from 0.2 to 0.6 h, the waiting time for drones almost doubles, but the total time cost increases by only 24%. The total time keeps stable, which proves that PDD is not sensitive to schedule changes. This is because the algorithm optimizes the waiting time spent by drone and takes the route with a short waiting time as the superior solution. If the drone has to wait for a long time for a bus, the algorithm will choose the transfer route with less time cost. As can be seen from the figure, our algorithm performs well at 0.4 h.

Fig. 13(a) and (b) shows that PDD outperforms the other two algorithms significantly in terms of time cost and distance cost of UAVs when the number of customers varies at [150, 450]. This is because the KGA and AMP algorithms are not fully applicable to our scenario, and it is inevitable that PDD performs better. The time cost of PDD is better than AMP for about 28% on average. As for the distance cost, PDD outperforms KGA by about 25.4% on average. With the increase in the number of customers, both the time cost and distance cost of PDD grow more slowly than others, which demonstrates that PDD has the highest efficiency among the three algorithms. The explanation is that the PDD algorithm uses multiple drones on buses to collaboratively deliver packages, jointly optimizing the time and distance costs of the

drones, as evidenced by the experimental results. Fig. 13(c) only has comparison results between KGA and PDD, which is due to the fact that AMP is applied to the VRPMT problem with only one type of vehicle, and the same is true for Fig. 14(c) below. When the number of customers is less than 225, the vehicle distance cost of KGA is slightly less than PDD. However, PDD outperforms KGA when the number of customers is large and the distance cost of PDD grows more slowly. This is because PTSs have multiple routes and stations, which provide flexibility for PDD to construct routes of the UAVs on the buses, i.e., PDDs can choose the optimal UAV drop-off point based on the route length, the waiting time, and other conditions, and balance the UAV flight distance and the distance of taking buses. From Fig. 13, we can conclude that PDD performs better in most cases.

The details shown in Fig. 14 depict the effect of UAV payload capacity on the three algorithms. It is obvious that PDD has the lowest time and distance cost, which means that PDD outperforms other algorithms. Fig. 14(a) and (b) shows that the tendency of PDD remains flat as the UAV payload capacity increases, which suggests that PDD is not sensitive to changes in payload capacity. Since the flight area of the UAV in PDD is limited by both payload capacity and power. Therefore, even with a large payload increase, the number of customers served by the UAV per trip will not increase because the power is limited. From Fig. 14(c), we can see that when the payload exceeds 3.5 kg, the distance cost corresponding to PDD increases slightly. In fact, UAVs use lighter and smaller batteries to save space for the payload, but at the same time shorten their own maximum flight distance. The limitation of flying distance causes UAVs to take public vehicles to go farther, and this slightly increases the distance cost of the vehicles. Thus, according to Fig. 14, PDD gets the best performance in time cost and distance cost compared with KGA and AMP.

VI. CONCLUSION

In this article, we have employed a PTS-assisted multidrone parcel delivery approach to overcome the limitations of the battery capacity and payloads of UAVs. To improve the efficiency of PTS-UAV collaboration, we have proposed a novel algorithm, referred to as PDD. PDD constructs and optimizes the routes of drones taking public vehicles as well as their flight routes, considering the impact of PTS schedules. The

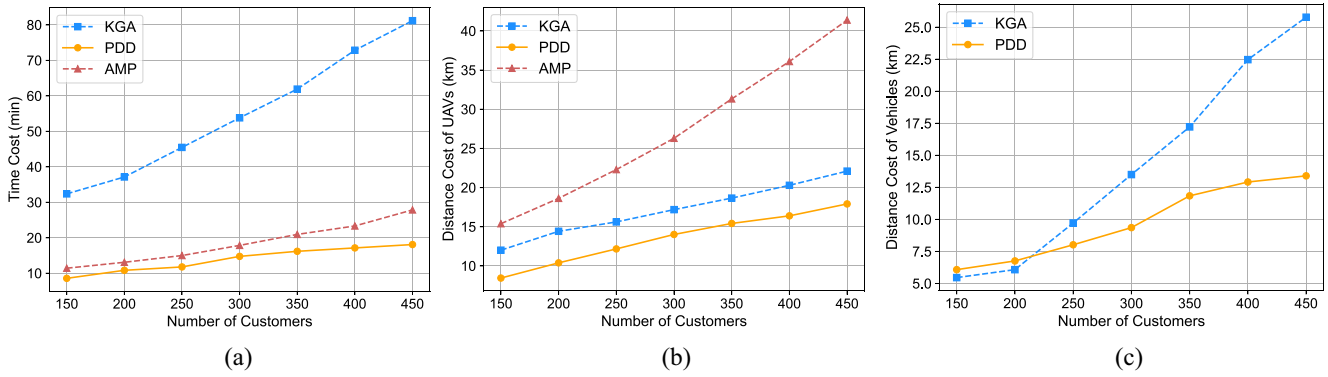


Fig. 13. Results with varying the number of customers. (a) Time cost. (b) Travel distance of UAVs. (c) Travel distance of vehicles.

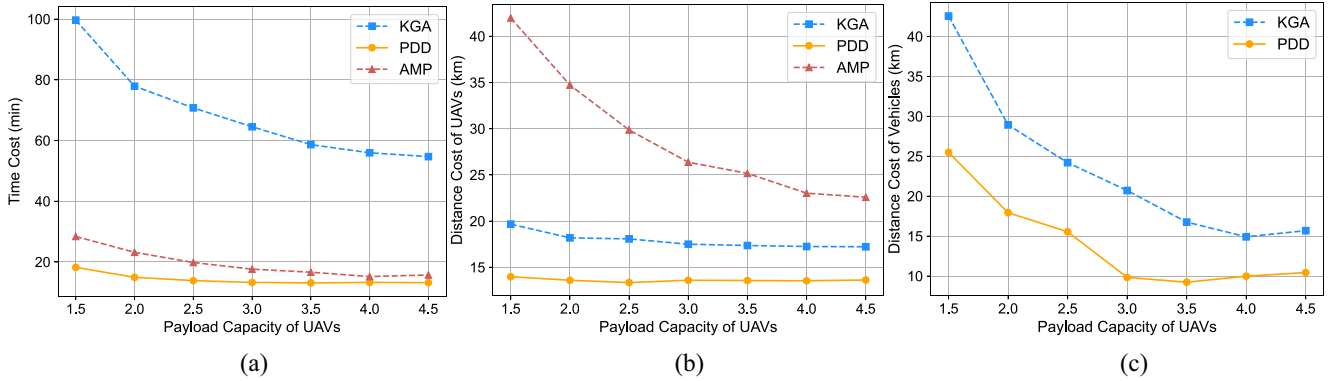


Fig. 14. Results with varying payload capacity of UAVs. (a) Time cost. (b) Travel distance of UAVs. (c) Travel distance of vehicles.

construction and optimization of the solution consist of the following three steps: 1) clustering customers; 2) planning UAV flight routes; and 3) planning and optimizing global routes, where the optimization is achieved by inserting and removing the existing routes. In addition, PDD optimizes the drone waiting time to conquer public vehicle travel limitations. Therefore, it has more application value. Through thorough experiments, the results have shown that PDD is not sensitive to schedule changes and makes the multidrone parcel delivery more efficient.

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